STATS 200A Lecture 7

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Theorem 7.1

Markov: $X \geq 0, a > 0 : \mathbb{P}(X \geq a) \leq \mathbb{E}(X)/a$
Chebyshev: $\mathbb{E}(X^2) < \infty, a > 0 : \mathbb{P}(|X - \mathbb{E}(X)| \geq a) \leq \mathrm{Var}(X)/a^2$

Example 7.2

$$\mathbb{E}\left(a\mathbf{1}\{X \geq a\} \leq X\right) \leq \mathbb{E}(X)$$

Solution.

$$\mathbb{E}(a\mathbf{1}\{X \geq a\}) = a\mathbb{E}(\mathbf{1}\{X \geq a\}) = a\mathbb{P}(X \geq n) \leq \mathbb{E}(X)$$
$$\mathrm{Var}(X) = \mathbb{E}(X - \mathbb{E}(X))^2$$

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Remark 7.3 (Chernoff Bound)

$$\mathbb{P}(X \geq a) \leq e^{-\theta a}\mathbb{E}\left[e^{\theta X}\right]$$

Theorem 7.4 (Jensen's Law)

$$\phi : \mathbb{R} \to \mathbb{R}, X \in L^1, \phi(X) \in L^1, \phi \geq 0, \phi(\mathbb{E}(X)) \leq \mathbb{E}(\phi(X))$$

Definition 7.5

For measurable f and $1 \leq p < \infty, \|f\|_p = \left(\int |f|^p \, d\mu\right)^{1/p}; \|f\|_\infty = \operatorname{ess\,sup} |f|$. $L^p(\mu) = \{f : \|f\|_p < \infty\}$. Identifying functions equal almost everywhere

Remark 7.6 (On a Probability Space)

For $1 \leq p \leq q \leq \infty, \|X\|_p \leq \|X\|_q \to L^\infty \subseteq L^2 \subseteq L^1$

$$X \in L^2(\mathbb{P}) \Leftrightarrow \mathbb{E}\,|X|^2 < \infty \Leftrightarrow \int |X|^2 \, d\mathbb{P} < \infty = \int x^2 \, d\mathbb{P}_X(x) = \int x^2 \, \mathbb{P}_X(dx)$$

$$X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$Y : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$\|X - Y\|_2^2 = \mathbb{E}\,|X - Y|^2 = \int |X - Y|^2 \, d\mathbb{P} = \int (x - y)^2 \, \mathbb{P}_{x,y}(dx, dy) = \int (x - y)^2 \, d\mathbb{P}_{X,Y}(x, y)$$

$$\int \sum_{i=1}^{\infty} f_i \, d\nu = \sum_{i=1}^{\infty} \int f_i \, d\nu$$

$$\mu(A) = \int_A f \, d\nu = \int f \cdot \mathbf{1}_A \, d\nu$$

$$\mu\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Definition 7.7

$\mu << \nu$ (absolutely continuous) if $\nu(A) = 0 \to \mu(A) = 0$

Theorem 7.8 (Radon-Nikodym)

μ, ν, σ-finite $\mu << \nu \to \mu = f\nu$ for a measurable $f \geq 0$, unique ν almost everywhere. Write $f = \frac{d\mu}{d\nu}$.